

10.5 Equivalent Length Of Columns :

In general we can write the Euler load $P_E = \frac{\pi^2 EI}{Cl^2}$ with given end conditions

equivalent length is the length of an equivalent column of the same material and section with hinged ends having the value of buckling load equal to that of the given column.

Where C is a constant which takes into account of the end conditions. If

- a) Column is hinged at both ends , $C = 1$
- b) Column is fixed at one end free at other , $C = 4$
- c) Both ends are fixed , $C = 1/4$
- d) One end is fixed and other is hinged , $C = 1/2$

In the place of using actual length of the column we can use equivalent length L_E

$$P_E = \frac{\pi^2 EI}{l_e^2} \quad \dots (10.15)$$

$$l_E = l\sqrt{C} \text{ known as equivalent length}$$

Table 1. Equivalent Length L_e Of Column Corresponding To Different End Conditions.

End Cond.	Both end hinged	One fix other free	Both fix	One fix other hinged
L_e	L	$2l$	$\frac{l}{2}$	$\frac{l}{\sqrt{2}}$

I is the moment of inertia of section about the axis of least resistance. Putting $I = Ak^2$. k is least radius of gyration.

$$P_E = \frac{\pi^2 E Ak^2}{l_e^2}$$

$$\frac{P_E}{A} = \frac{\pi^2 E}{(l/K)^2} \quad \dots\dots\dots(10.16)$$

The ratio (l/k) is the slenderness ratio for the column Euler formula is applicable only if l/k is greater than 80.4 for steel column hinged at both ends.

10.6 Limitation of Euler's Formula:

$$P_{CR} = \frac{\pi^2 EI}{L_e^2} = \frac{\pi^2 EAK^2}{l_e^2}$$

$$\text{Crippling stress } \sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 E}{(l_e/k)^2}$$

Where l/k is slenderness ratio see figure 10.6. If the (l/k) is small the crippling stress (or stress at failure) will be high but for the column material the σ_{cr} can not be greater than crushing stress

(Proportional limit). Hence when the slenderness ratio is less than a certain limit, Euler formula gives a value greater than the crushing stress.

In the limiting case we can find the value of (l/k) for which crippling stress is equal to crushing stress. For mild steel column with both ends hinged.

$$\text{Crushing stress} = 330 \text{ N/mm}^2$$

$$\frac{\pi^2 EI}{(l/k)^2} = 330 \quad E = 2 \times 10^5 \text{ N/mm}^2$$

$$\frac{l}{k} = \sqrt{\frac{\pi^2 EI}{330}} = \sqrt{6282} = 79.27 \approx 80$$

Hence for $(l/k) < 80$ for mild steel Euler's formula is not valid.

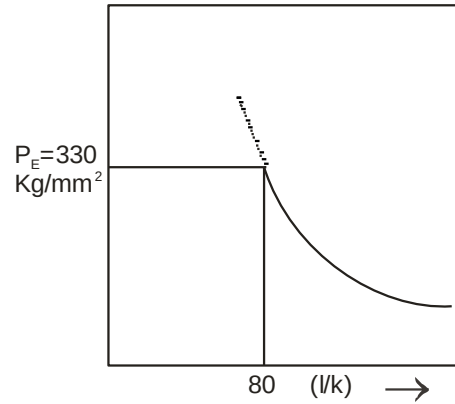


Fig. 10.7

10.7 Empirical Relations :

Euler formula does not take into account the direct compressive stress and hence give correct results only for long columns. For medium struts or columns Euler's formula does not give correct results. The following empirical formulas are important from the subject point of view.

10.7.1 Rankine's Formula For Columns :

Rankine proposed an empirical formula for columns which covers all cases of columns ranging from very short to long struts. He proposed the relation

$$\frac{1}{P} = \frac{1}{P_C} + \frac{1}{P_E} \quad \dots(10.17)$$

Where $P_C = \sigma_c A$ (crushing load)

$$P_E = \frac{\pi^2 EI}{l^2} \quad (\text{Euler crippling load}).$$

In equation (10.17) $\frac{1}{P_C}$ is constant for a material. For short column P_E is large and hence

$\frac{1}{P_E}$ is small as compared to $\frac{1}{P_C}$.

$$\text{Hence } \frac{1}{P} \cong \frac{1}{P_C} \text{ or } P \cong P_C.$$

Similarly for long column P_E is small and $\frac{1}{P_E}$ is much large as compared to $\frac{1}{P_C}$.

$$\text{Hence } \frac{1}{P} \cong \frac{1}{P_E}, \quad P \cong P_E.$$

Thus value of p obtained from above relation covers extreme cases of long as well as short columns. Hence Rankine formula is taken to be valid for length of columns.

$$\text{Hence } P = \frac{P_C P_E}{P_E + P_C} = \frac{1}{\frac{1}{P_E} + \frac{1}{P_C}} = \frac{1}{\frac{1}{\frac{\sigma_c A}{\pi^2 EI}} + \frac{1}{\sigma_c A}}$$

Putting $I = AK^2$ where K is least radius of gyration of column section.

$$P = \frac{\sigma_c A}{1 + \frac{\sigma_c}{\pi^2 E} \left(\frac{l}{K} \right)^2} = \frac{\sigma_{Cr} A}{1 + a \left(\frac{l}{K} \right)^2} \quad \dots(10.18)$$

σ_c – Crushing stress for the metal

a – Rankine constant for metal = $\frac{\sigma_{Cr}}{\pi^2 E}$

Value of σ_c and a are shown in table 2

Table 2

No.	Material	σ_c (N/mm ²)	a (for ends hinged)
1.	Wrought Iron	250	1/9000

2.	CI	550	1/1600
3.	MS	320	1/7500
4.	Strong Timber	50	1/750

For all condition of column

$$P = \frac{\sigma_c A}{1 + a \left(\frac{L_e}{K} \right)^2} = \frac{\text{Crushing load}}{1 + a \left(\frac{L_e}{K} \right)^2}$$

$1 + a \left(\frac{L_e}{K} \right)^2$ is the factor takes into account the buckling effect.

10.7.2 Johnson's Formula For Column :

Euler's formulae and Rankine formula are not exact compared with actual result obtained by experiments because

- Effect of direct compression is neglected in Euler's Formula.
- The loading is not exactly applied as desired.
- The hinge joints are not perfectly frictionless.
- The fixed ends are not absolutely fixed.

10.7.2.1 Johnson's Straight Line Formula For Columns

$$P = A \left[\sigma_c - n \left(\frac{L}{K} \right) \right] \quad \dots(10.19)$$

Where P = Safe load on the column

n = constant whose value depends upon column material (Table 3)

Table 3

S.no.	Material	n	r
1.	M.S.	0.0053	0.000057
2.	Wrought Iron	0.0053	0.000039
3.	CI	0.008	0.000016

10.7.2.2 : Johnson's Parabolic Formula For Columns

$$P = A \left[\sigma_c - r \left(\frac{L}{K} \right) \right] \quad \dots(10.20)$$

$$\frac{P}{A} = \frac{\sigma_{CA}}{1 + a \left(\frac{Le}{k} \right)^2} \quad \text{Constant given in Table 3} \quad \dots(10.21)$$

This formula involves factor of safety. σ_{CA} is allowable crushing stress.

10.8 Long Columns Subjected To Eccentric Loading :

In actual practice load is eccentric. Direct compression stress

$$\sigma_C = \frac{P}{A}$$

Let P = load on column

A – Area of cross section

e – Eccentricity

z – Section modulus

k – Least radius of gyration

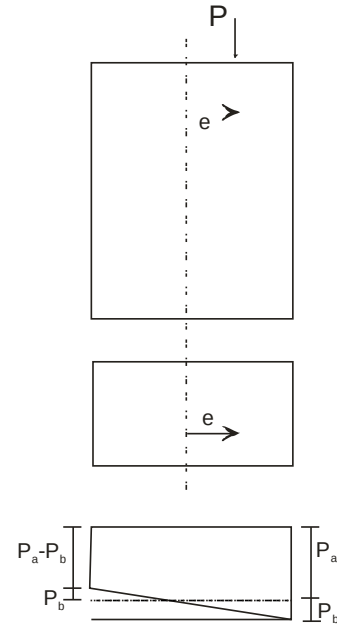


Fig. : 10.8

Bending couple will cause longitudinal tensile and

compressive stresses. The fiber stress σ_b at any distance y_c from neutral axis NA is given

by

$$\sigma_b = \frac{My_c}{I} = \frac{P_e y_c}{I_x}$$

Where y_e – distance of extreme fiber (on comp. side) from the axis of the column.

Maximum stress at any section is given by

$$\sigma_{\max} = \frac{P}{A} + \frac{P_e y_c}{I_x} = \frac{P}{A} + \frac{M}{Z} \quad \dots(10.22)$$

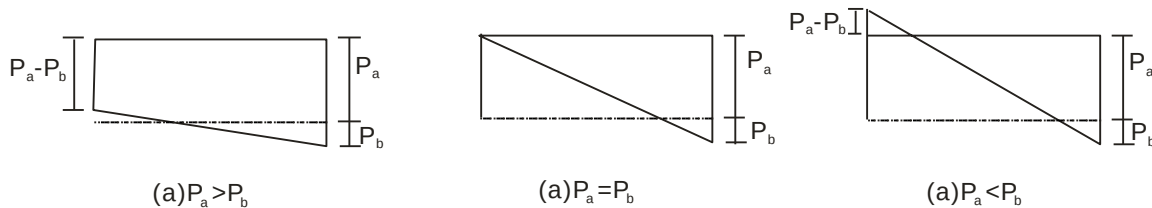


Fig. : 10.9

$$\sigma_{\max} = \frac{P}{A} + \frac{P_e y_c}{AK^2} \quad \left(z = \frac{I}{y_e} = \frac{AK^2}{y_e} \right)$$

$$\sigma_{\max} = \frac{P}{A} \left(1 + \frac{e y_c}{K^2} \right)$$

σ_c = permissible stress for the metal.

$$P = \frac{\sigma_c A}{\left(1 + \frac{e y_c}{K^2} \right)} \quad \dots(10.23)$$

$1 + \frac{e y_c}{K^2}$ is the reduction factor which take into account the eccentricity of loading. If the effect of buckling is also to be taken into account the safe axial load with eccentricity is given by.

$$P = \frac{A}{\left(1 + \frac{e y_c}{K^2} \right) \left(1 + a \frac{l^2}{K^2} \right)} \quad \dots(10.24)$$

10.9 The Euler Secant Formula :

Let us consider a long column AB of length l and both ends hinged loaded with force P . At any section from distance x , y is the deflection with reference to the line of action of force P , which is acting at an eccentricity e at the ends.

The BM due to load is $-Py$.

Hence $EI \frac{d^2 y}{dx^2} = -Py$

$$\text{or } \frac{d^2 y}{dx^2} + \frac{P}{EI} y = 0$$

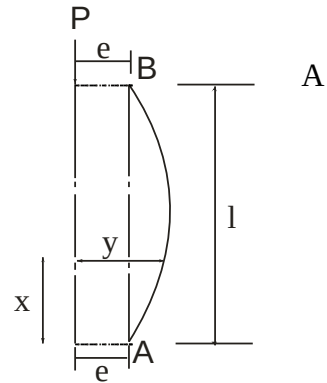


Fig.10.10

Solution of the above differential equation is given by

$$y = C_2 \cos x \sqrt{\frac{P}{EI}} + C_1 \sin x \sqrt{\frac{P}{EI}} \quad \dots(10.25)$$

C_1 and C_2 are constants found by applying boundary conditions.

$$\text{At } x = 0, y = e \quad \therefore C_2 = e$$

$$\frac{dy}{dx} = -C_2 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_1 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}}$$

$$\text{at } x = \frac{l}{2} \text{ (mid of column)}$$

$$\text{slope } \frac{dy}{dx} = 0$$

$$0 = -e \sqrt{\frac{P}{EI}} \sin \frac{l}{2} \sqrt{\frac{P}{EI}} + C_1 \sqrt{\frac{P}{EI}} \cos \frac{l}{2} \sqrt{\frac{P}{EI}}$$

$$C_1 = e \left[\frac{\sin \frac{l}{2} \sqrt{\frac{P}{EI}}}{\cos \frac{l}{2} \sqrt{\frac{P}{EI}}} \sin x \sqrt{\frac{P}{EI}} + \cos x \sqrt{\frac{P}{EI}} \right] \quad \dots(10.26)$$

$$x = l/2 \quad y_{\max} = e \left[\frac{\sin^2 \frac{l}{2} \sqrt{\frac{P}{EI}}}{\cos \frac{l}{2} \sqrt{\frac{P}{EI}}} + \cos \frac{l}{2} \sqrt{\frac{P}{EI}} \right] = e \sec \frac{l}{2} \sqrt{\frac{P}{EI}} \quad \dots(10.27)$$

$$\text{Maximum bending moment} = P y_{\max} = P e \sec \frac{l}{2} \sqrt{\frac{P}{EI}}$$

Maximum compressive stress

$$\sigma_{\max} = \frac{P}{A} + \frac{M}{Z}$$

$$= \frac{P}{A} + \frac{P_e \sec \frac{l}{2} \sqrt{\frac{P}{EI}}}{Z} \quad \left(Z = \frac{AK^2}{y_c} \right)$$

$$= \frac{P}{A} \left[1 + \frac{e y_c}{K^2} \sec \frac{l}{2} \sqrt{\frac{P}{EI}} \right] \quad \dots(10.28)$$

This known s Euler Secant formula for columns for all columns for all cases

$$\sigma = \frac{P}{A} + \frac{P_e \sec \frac{LE}{Z} \sqrt{\frac{P}{EI}}}{Z} \quad \dots(10.29)$$

It can be easily proved that if one end is fixed and the other end is free, the maximum bending moment is $= P e \sec l \sqrt{\frac{P}{EI}}$

Similarly if both ends are fixed maximum bending moment $= P.e . \sec \frac{l}{4} \sqrt{\frac{P}{EI}} .$